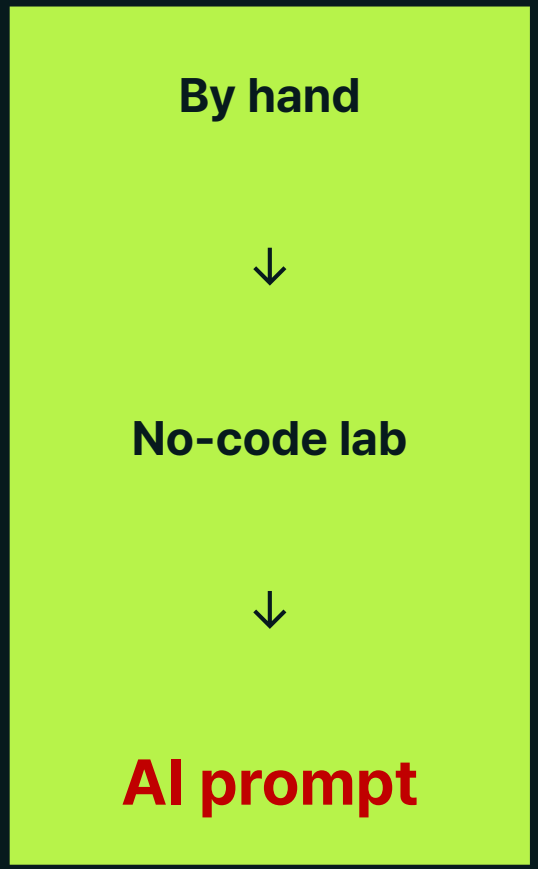
Honest Backtesting Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 10 Workbook

Monthly Sharpe 0.5 is about 1.73 annualized

Monthly excess return 1%, volatility 2%, i.i.d. stable variance

Step 1 · Monthly Sharpe, then annualize

$$SR_{\text{monthly}} = \frac{1\%}{2\%} = 0.5, \qquad SR_{\text{annual}} \approx 0.5 \times \sqrt{12} = 0.5 \times 3.464 \approx 1.73$$

Step 2 · Run 1,000 tests at the 5% level

$1000 \times 0.05 = 50$ significant by chance

Strong autocorrelation breaks simple annualization

The best of 1,000 no-skill strategies has Sharpe about 2.15

3 years, 756 daily obs., N=1,000

Step 1 · Source formula

Approx. best Sharpe by luck $\approx \sqrt{252}\,\dfrac{\sqrt{2\ln N}}{\sqrt{T}}$, N = number of trials, T = number of daily obs

Step 2 · Plug in

$$\sqrt{252}\,\frac{\sqrt{2\ln 1000}}{\sqrt{756}} = 15.87 \times \frac{\sqrt{13.82}}{27.50} = 15.87 \times \frac{3.717}{27.50} \approx 2.15$$

This is neither the DSR nor a strict significance bound

Shading label windows shows which samples to purge

Starts on days 1–10, 3-day return labels, test days 5–7

1. Where

Draw date cells on paper or a notepad.

2. Input

Write each sample's start and 3-day label span.

3. Do and output

Shade test dates; cross out overlapping training samples.

4. Check

Note if endpoints count and if post-test data is allowed.

Prompt for the AI chat

I am a non-major. Tabulate start and end dates of 10 samples with 3-day return labels; the test is days 5–7. Using closed intervals, mark overlapping training samples. Compare walk-forward with purged CV (training on both sides) and explain how future-data use differs from label overlap.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.